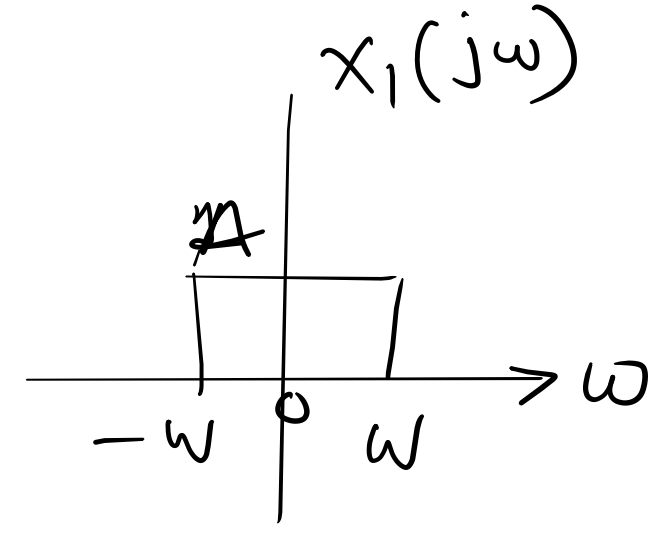
4.21(f) $x(t) = 2\left(\frac{\sin \pi t}{\pi t}\right)\left(\frac{\sin 2\pi(t-1)}{2\pi(t-1)}\right)$

$= x_1(t)\ x_2(t)$



$x_1(t)$

$x_2'(t) = \frac{\sin 2\pi t}{2\pi t}$

$X_1(j\omega)$

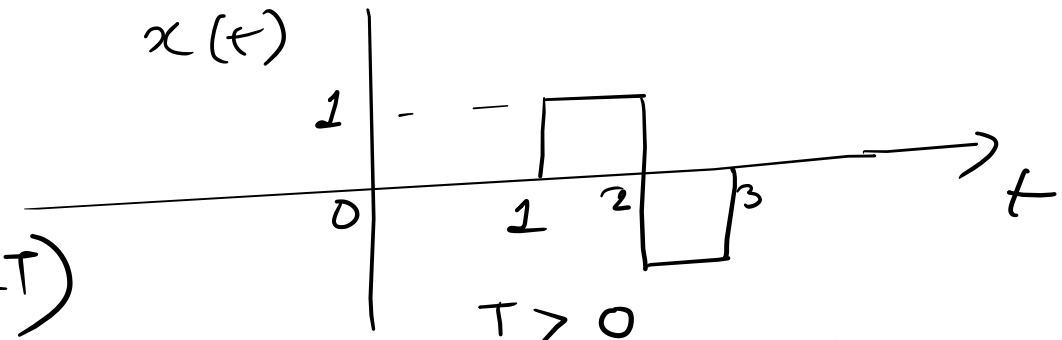
$x_2(t) = x_2'(t-1)$

$X_2(j\omega)$

$X(j\omega) = \frac{1}{2\pi}\{X_1(j\omega) * X_2(j\omega)\}$

4.27

$$x(t) = u(t-1) - 2u(t-2) + u(t-3)$$



$X(j\omega)$

$$\left.\frac{X(j\omega)}{T}\right|_{\omega = \frac{2\pi k}{T}} = a_k$$

$= \omega_0 k$

$\tilde{x}(t)$ periodic signal

$$\tilde{x}(t) = \sum_{k=-\infty}^{\infty} x(t - kT)$$

$$= \sum_{k=-\infty}^{\infty} a_k e^{jk\frac{2\pi}{T}t}$$

$$a_k = \frac{1}{T} X\left(j\frac{2\pi k}{T}\right)$$

Time — Freq.

$x(t) \rightleftharpoons X(j\omega)$

$\tilde{x}(t) \rightleftharpoons a_k$

F.S.

$$\tilde{x}(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\frac{2\pi}{T}t}$$

$$a_k = \frac{1}{T}\int_T \tilde{x}(t)\, e^{-jk\frac{2\pi}{T}t}\, dt$$

$\equiv$

Over one time period $\tilde{x}(t) = x(t)$

$$a_k = \frac{1}{T}\int_{-\infty}^{\infty} x(t)\, e^{-jk\frac{2\pi}{T}t}\, dt = \frac{1}{T} X(j\omega)\Big|_{\omega = k\frac{2\pi}{T}}$$

$\therefore$ $x(t)$ is zero beyond that one time period —

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)\, e^{j\omega t}\, d\omega$$

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t}\, dt$$

$\equiv$

D.T. Fourier Transform

$$x[n] \rightleftharpoons X(e^{j\omega})$$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]\, e^{-j\omega n}$$

$$x[n] = \frac{1}{2\pi}\int_{2\pi} X(e^{j\omega})\, e^{j\omega n}\, d\omega$$

C.T. F.T.

$$x(t) \rightleftharpoons X(j\omega)$$

$\frac{d\phi}{dt} = \omega$

$\cos(\omega t)$

$\phi = \omega t$

$\phi = \omega n$

$\Rightarrow \phi$

Ex

Case - 1

$x[n] = a^n u[n] \quad |a| < 1$

$0 < a < 1$

$x[n] = a^n u[n]$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

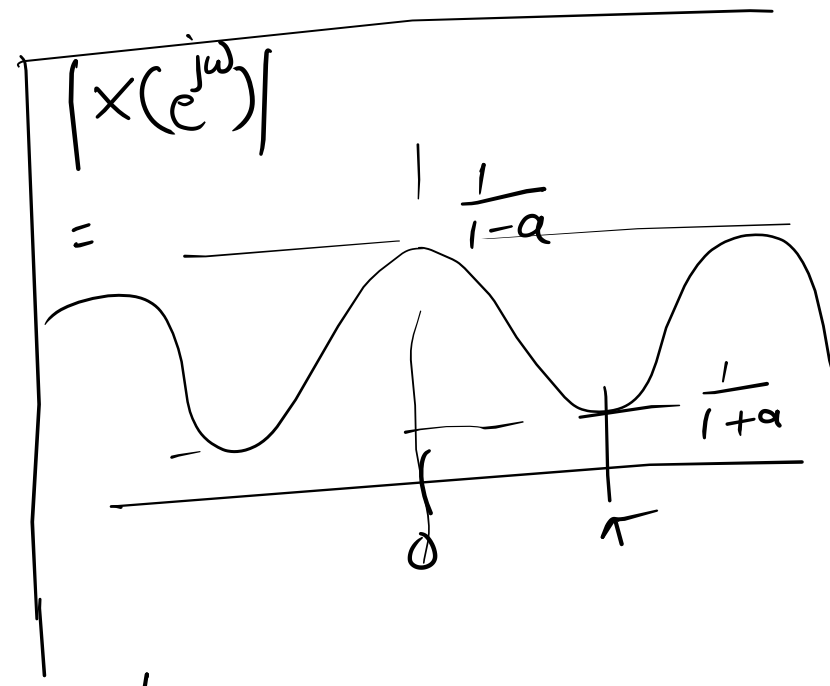
$$= \sum_{n=0}^{\infty} a^n e^{-j\omega n}$$

$$= \sum_{n=0}^{\infty} (a e^{-j\omega})^n = \frac{1}{1 - a e^{-j\omega}}$$

$\omega = 0 \Rightarrow X(e^{j\omega}) = \frac{1}{1-a}$

$\omega = \pi \Rightarrow X(e^{j\omega}) = \frac{1}{1+a}$

$|\omega| \leq \pi$



Case-2 $\quad -1 < a < 0$

$$x[n] = a^n u[n]$$

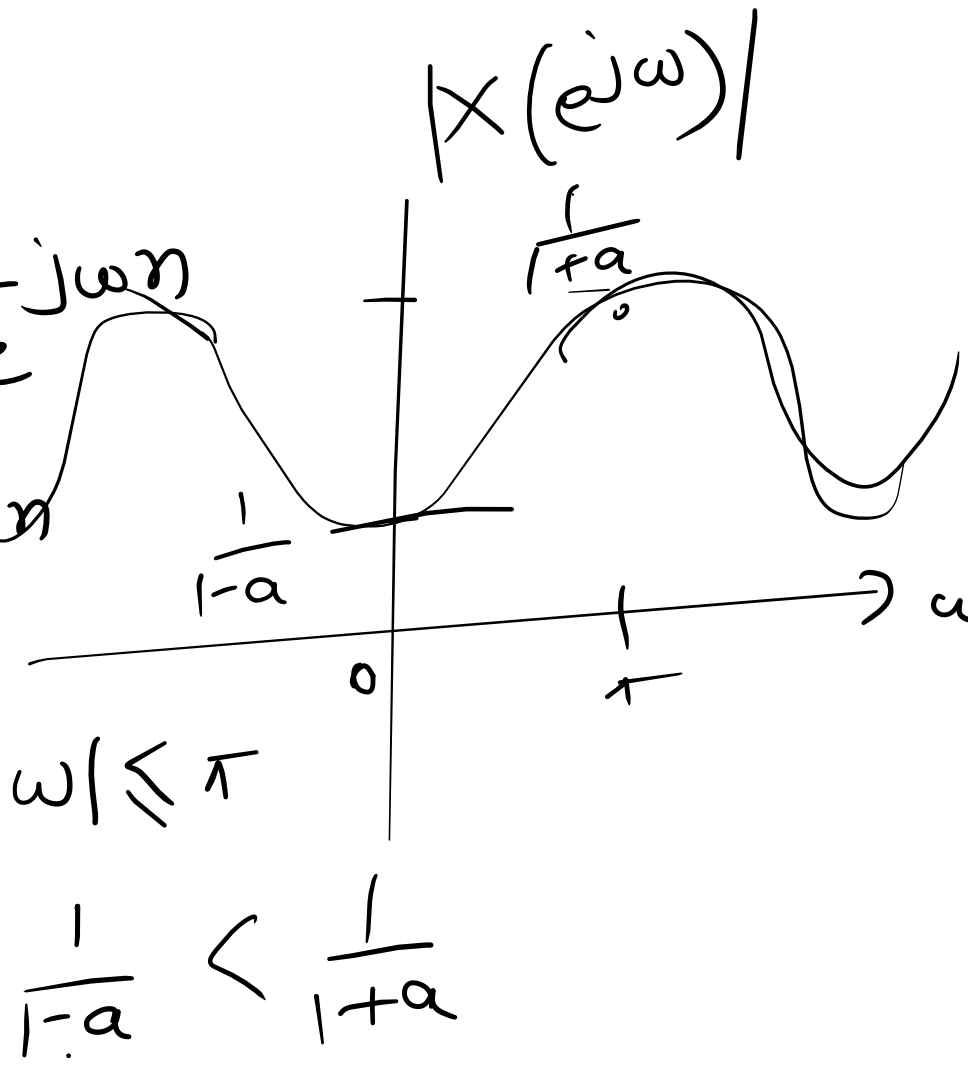
$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

$$= \sum_{n=0}^{\infty} a^n e^{-j\omega n}$$

$$= \frac{1}{1 - a e^{-j\omega}} \; ; \quad |\omega| \le \pi$$

$\omega = 0 \implies \frac{1}{1-a}$

$\omega = \pi \implies \frac{1}{1+a}$



$$\frac{1}{1-a} < \frac{1}{1+a}$$

Ex $x[n] = a^{|n|}$ $\boxed{0<a<1}$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{-1} a^{-n} e^{-j\omega n} + \sum_{n=0}^{\infty} a^{n} e^{-j\omega n}$$

$$= \sum_{n=\infty}^{1} a^{n} e^{j\omega n} + \frac{1}{1-ae^{-j\omega}}$$

$$= \sum_{n=1}^{\infty} a^{n} e^{j\omega n} + \frac{1}{1-ae^{-j\omega}}$$

$$= ae^{j\omega} + (ae^{j\omega})^2 + \cdots + \frac{1}{1-ae^{-j\omega}}$$

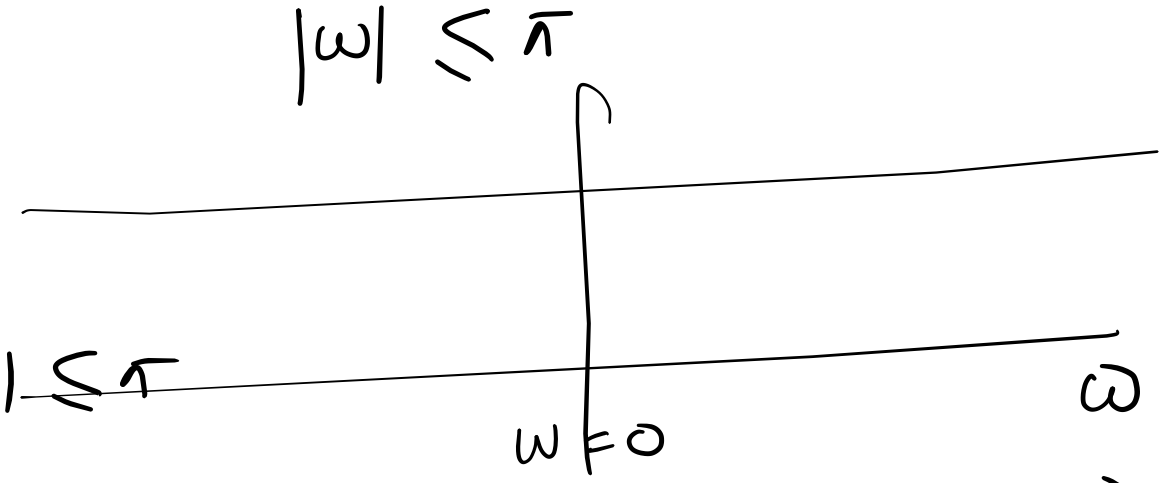
$$= ae^{j\omega}\left[1 + ae^{j\omega} + (ae^{j\omega})^2 + \cdots\right] + \frac{1}{1-ae^{-j\omega}}$$

$$X(e^{j\omega}) = \frac{ae^{j\omega}}{1-ae^{j\omega}} + \frac{1}{1-ae^{-j\omega}}$$

Ex $x[n] = \delta[n]$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

$$= 1$$

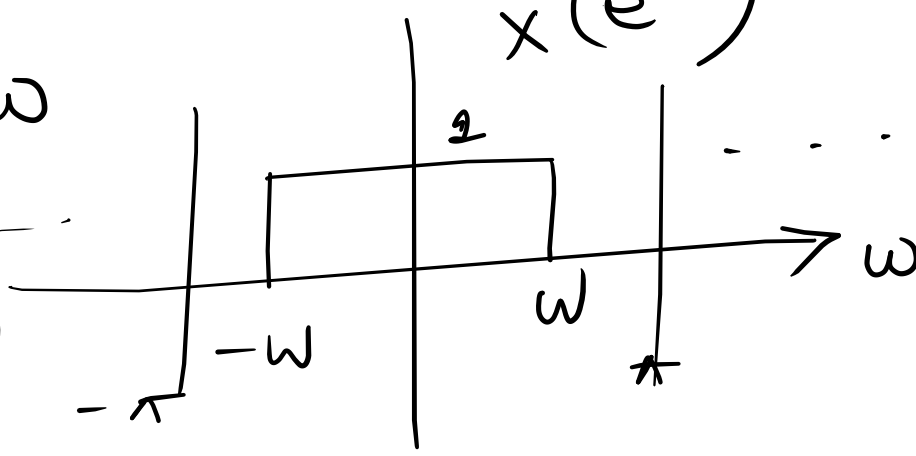


Given $X(e^{j\omega}) = 1$ $|\omega| \le \pi$

Find $x[n] = ?$

$$x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

Let $\hat{x}[n] = \frac{1}{2\pi} \int_{-W}^{W} \hat{X}(e^{j\omega}) e^{j\omega n} d\omega$



$$\hat{x}[n] = \frac{\sin Wn}{\pi n}$$

$W \longrightarrow \pi$

$$\begin{cases} \text{For all } n \neq 0, & \hat{x}[n] = 0 \\ \text{For } n = 0, & \hat{x}[n] = 1 \end{cases}$$

$W \longrightarrow \pi$

$\hat{x}[n] \longrightarrow \delta[n] = x[n]$

C.T.F.T.

periodic signal $x(t)$ ⟶ F.S. expansion

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

$$X(j\omega) = \sum_{k=-\infty}^{\infty} 2\pi a_k \delta(\omega - k\omega_0)$$

$1 \rightleftharpoons 2\pi\delta(\omega)$

$\delta(t) \rightleftharpoons 1$

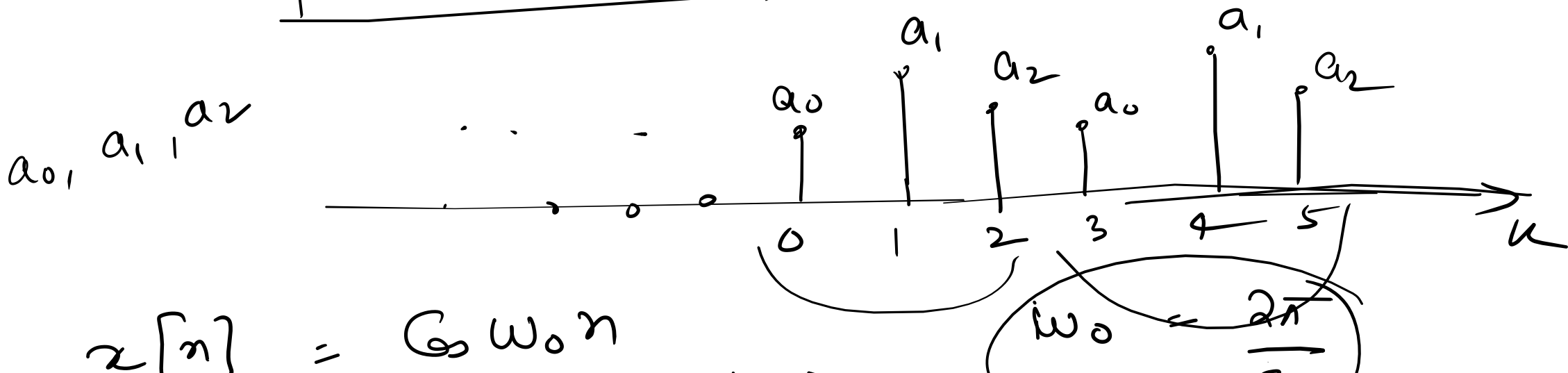
$e^{j\omega_0 t} \rightleftharpoons 2\pi\delta(\omega-\omega_0)$

$e^{jk\omega_0 t} \rightleftharpoons 2\pi\delta(\omega-k\omega_0)$

D.T. F.T.

Given a periodic signal $x[n]$

$$x[n] = \sum_{k=<N>} a_k e^{jk\omega_0 n} \qquad \omega_0 = \frac{2\pi}{N}$$



Ex $x[n] = \cos \omega_0 n$

$\omega_0 < \frac{2\pi}{3}$

periodic signal

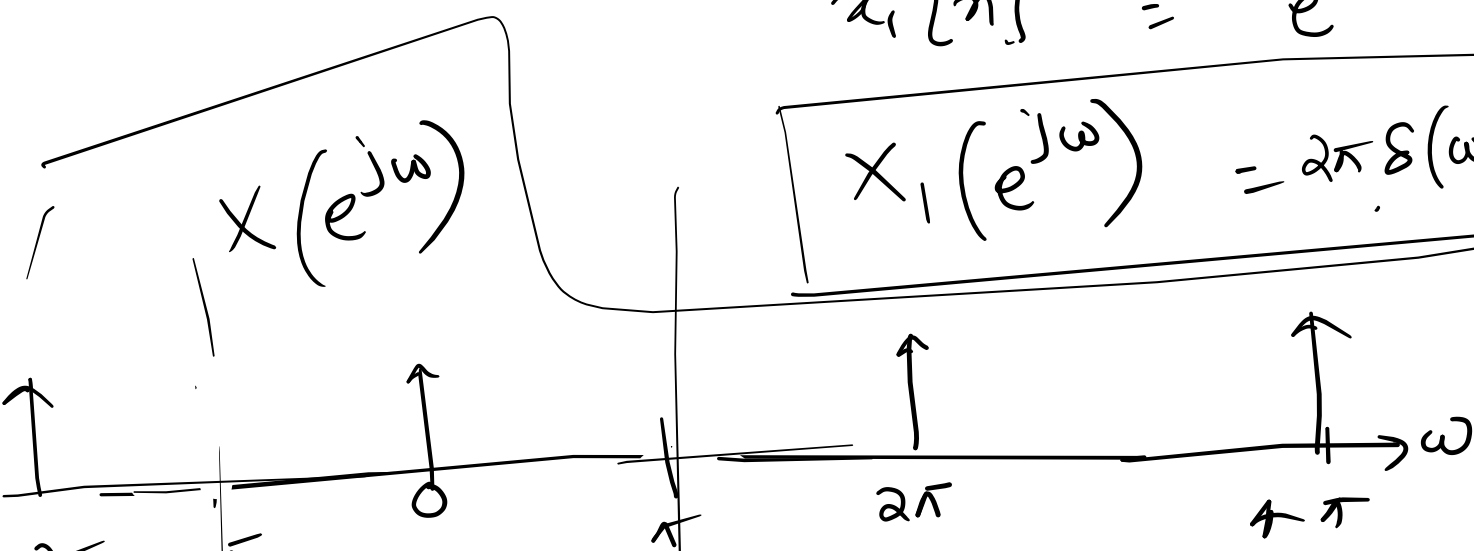
$$= \frac{e^{j\omega_0 n} + e^{-j\omega_0 n}}{2} = \frac{1}{2} e^{j\omega_0 n} + \frac{1}{2} e^{-j\omega_0 n}$$

$x_1[n] = e^{j\omega_0 n}$ $\quad \omega_0 = \frac{2\pi}{3}$

$$X_1(e^{j\omega}) = 2\pi\delta(\omega-\omega_0), \quad |\omega| \le \pi$$

$\delta[n] \rightleftharpoons 1$

$1 \rightleftharpoons 2\pi\delta(\omega)$



$$x[n] = \frac{1}{2\pi}\int_{2\pi} 2\pi\delta(\omega)\, e^{j\omega n}\, d\omega$$

$$= \frac{2\pi}{2\pi} = 1$$

$$X(e^{j\omega}) = 2\pi\delta(\omega) \qquad |\omega| \le \pi$$

$e^{j\omega_0 n} \rightleftharpoons 2\pi\delta(\omega-\omega_0)$

$$x_1[n] = e^{j\omega_0 n}$$

$$X_1(e^{j\omega}) = 2\pi\,\delta(\omega - \omega_0)\,; \quad |\omega| \le \pi$$

$$\boxed{X_1(e^{j\omega}) = 2\pi \sum_{l=-\infty}^{\infty} \delta(\omega - \omega_0 + 2\pi l)}$$

$$x[n] = \cos\omega_0 n = \frac{1}{2} e^{j\omega_0 n} + \frac{1}{2} e^{-j\omega_0 n}$$

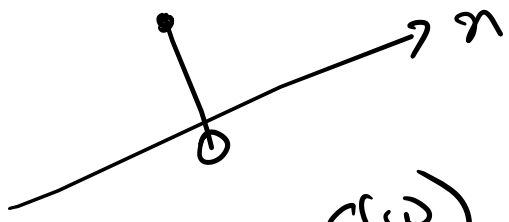
D.T.F.T. $$X(e^{j\omega}) = \frac{1}{2}\, 2\pi \sum_{l=-\infty}^{\infty} \delta(\omega - \omega_0 + 2\pi l)$$

$$+ \frac{1}{2} \cdot 2\pi \sum_{l=-\infty}^{\infty} \delta(\omega + \omega_0 + 2\pi l)$$

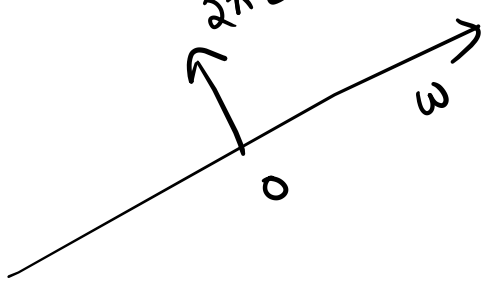
Or $X(e^{j\omega}) = \pi\delta(\omega-\omega_0) + \pi\delta(\omega+\omega_0)$ $|\omega| \le \pi$

$x[n] = \sum_{k=\langle N \rangle} a_k e^{jk\omega_0 n}$

$x[n] = \delta[n]$



$X(e^{j\omega}) = 2\pi\delta(\omega)$



↑ periodic signal with time period $N = 3$

$$x[n] = a_0 + a_1 e^{j\omega_0 n} + a_2 e^{j2\omega_0 n}$$

$$X(e^{j\omega}) = 2\pi a_0 \delta(\omega) + 2\pi a_1 \delta(\omega-\omega_0) + 2\pi a_2 \delta(\omega-2\omega_0)$$

$|\omega| \le \pi$

Properties —

① Linearity

$$x_1[n] \rightleftharpoons X_1(e^{j\omega})$$

$$x_2[n] \rightleftharpoons X_2(e^{j\omega})$$

$$a x_1[n] + b x_2[n] \rightleftharpoons a X_1(e^{j\omega}) + b X_2(e^{j\omega})$$

---

② Time Shifting

$$x[n] \rightleftharpoons X(e^{j\omega})$$

$$x[n-n_0] \rightleftharpoons e^{-j\omega n_0} X(e^{j\omega})$$

$$|X(e^{j\omega})|$$

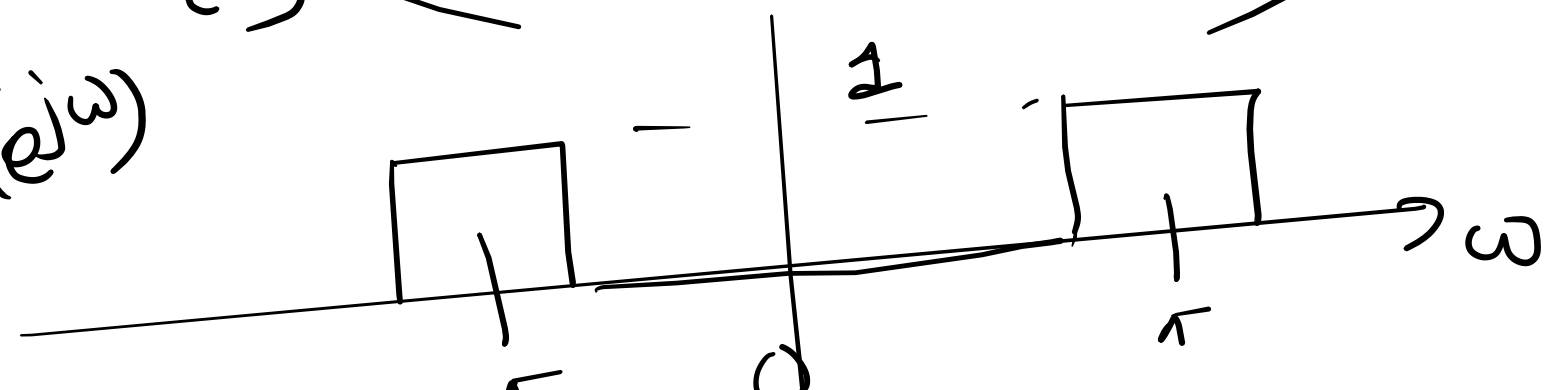
$$|e^{-j\omega n_0} X(e^{j\omega})|$$

(3) Shifting in the frequency domain

$$x[n] \rightleftharpoons X(e^{j\omega})$$

$$e^{j\omega_0 n}\, x[n] \rightleftharpoons X(e^{j(\omega-\omega_0)})$$

$h[n] \longrightarrow H(e^{j\omega})$



$0 \equiv$ lowest frequency

$\pi \equiv$ highest frequency

$e^{j\omega_0 n}$

$e^{j\pi n}$

$$h[n] \rightleftharpoons H(e^{j\omega})$$

$$(-1)^n h[n] \rightleftharpoons H(e^{j(\omega-\pi)})$$

$$h[n] \equiv \begin{bmatrix} 0 & -0.5 & 1 & -0.5 \end{bmatrix} \leftarrow$$

$$(-1)^n h[n] \equiv \underbrace{\begin{bmatrix} 0 & 0.5 & 1 & 0.5 \end{bmatrix}}$$

(4) Time reversal

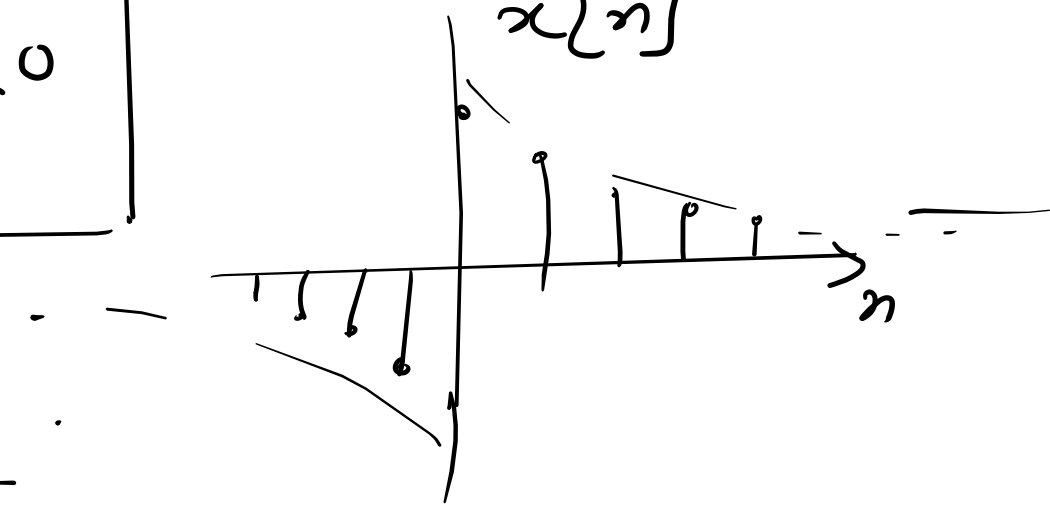
$$x[n] \rightleftharpoons X(e^{j\omega})$$

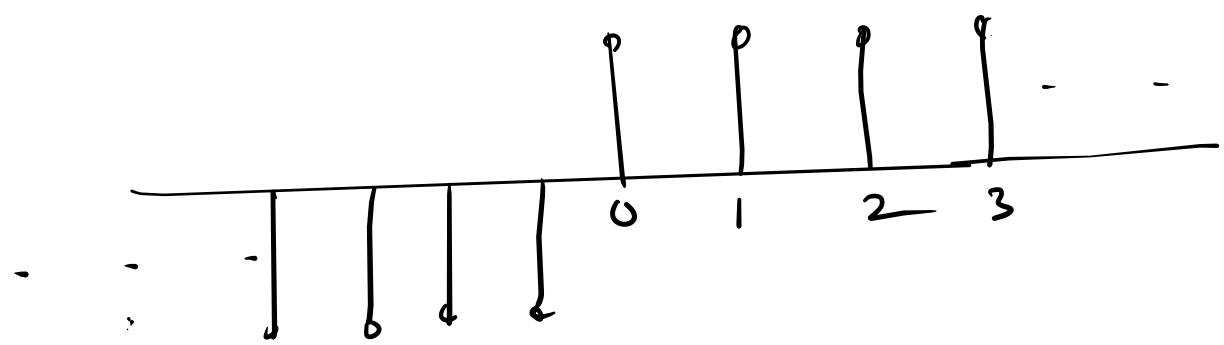
$$x[-n] \rightleftharpoons X(e^{-j\omega})$$

$$x[n] = \begin{cases} a^n & n \geq 0 \\ -a^{-n} & n < 0 \end{cases} \qquad 0 < a < 1$$

Ex

$a \to 1$



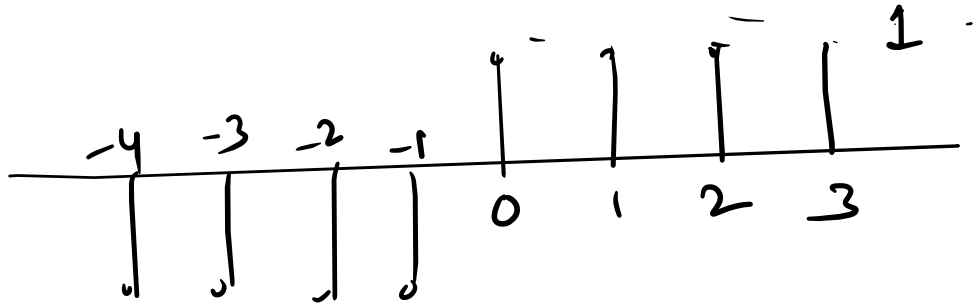


$$X(e^{j\omega}) = \sum_{n=0}^{\infty} a^n e^{-j\omega n} - \sum_{n=-\infty}^{-1} a^{-n} e^{-j\omega n}$$

$$= \frac{1}{1 - a e^{-j\omega}} - \frac{1}{1 - a e^{j\omega}} + 1$$

$$= \frac{1 + a^2 - 2a e^{j\omega}}{1 + a^2 - a e^{-j\omega} - a e^{j\omega}}$$

if $a \to 1$ $X(e^{j\omega}) = \frac{2(1 - e^{j\omega})}{(1 - e^{j\omega})(1 - e^{-j\omega})} = \frac{2}{1 - e^{-j\omega}}$



$$x[n] = \begin{cases} 1 & n \geq 0 \\ -1 & n < 0 \end{cases} \rightleftharpoons X(e^{j\omega}) = \frac{2}{1-e^{-j\omega}} \; ; \; |\omega| \leq \pi$$

$$u[n] = \frac{1}{2}\{x[n] + 1\}$$

$$U(e^{j\omega}) = \frac{1}{1-e^{-j\omega}} + \pi\delta(\omega) \; ; \; |\omega| \leq \pi$$

LTI System ; System is Accumulator $\equiv$ $h[n] = u[n]$

Multiplication in time domain ⟺ Convolution in frequency domain

convolution ⟺ multiplication

$$h[n] * x[n] \rightleftharpoons X(e^{j\omega})\,H(e^{j\omega})$$

if $h[n] = u[n]$

$$y[n] = \sum_{k=-\infty}^{n} x[k] \rightleftharpoons X(e^{j\omega})\,U(e^{j\omega})$$

$$= X(e^{j\omega})\left[\frac{1}{1-e^{-j\omega}} + \pi\delta(\omega)\right]; \quad |\omega| \le \pi$$

$$\boxed{Y(e^{j\omega}) = \frac{X(e^{j\omega})}{1-e^{-j\omega}} + \pi X(e^{j0})\delta(\omega)}; \quad |\omega| \le \pi$$